

STATEMENT OF WORK – ENERGY STORAGE MODULE TESTING

1. Background/ Test Objective:

The vast majority of the electric Vertical Take-off and Landing (eVTOL) aircraft currently in design or prototype stages utilize electric or hybrid electric propulsion systems. These systems consist of the Energy Storage Systems (ESS), which are typically large Lithium-Ion (Li-Ion) battery modules and associated battery management systems (BMS) connected to a variety of electric motors and propellers. This type of system is a new alternative to conventional liquid propulsion systems, and has been created to operate quietly and economically, which are considerations for operating in the urban environment, all while producing zero or close to zero emissions, which are environmental factors. One barrier hindering potential growth of these new aircraft is the airworthiness certification aspect for the ESS systems. The certification for the ESS systems will have to be based on a variety of requirements, one of which may involve the survivability of the ESS in a dynamic mishap event or crash. The performance of the ESS systems undergoing loading within the dynamic impact environment is unknown, resulting in a gap in the current knowledge base. This barrier will need to be overcome before widespread adaptation can occur.

To address this gap, NASA is engaging in a test program to dynamically drop test NASA provided ESS test hardware in order to:

- Determine the applicability of a 50 ft drop test, utilizing existing FAA regulations.
- Determine the performance and survivability of the ESS to a 50 ft drop test.
- Generate data to support regulations development

2. Requirements/Duties:

In order to fulfill the test objective mentioned above, test lab shall perform all of the following duties related to the dynamic drop testing:

Testing Services:

- Conduct 1 drop test of NASA supplied ESS pack test articles along with one practice test utilizing a drop tower or other mechanism capable of lifting and dropping up to 1,000 lb on a drop trolley/platform from heights up to 50 ft utilizing a guided system capable of near free fall flight with controlling impact orientations. Test articles may be configured to include a mounting surface and energy absorbing material present.
- Provide temporary storage of NASA ESS pack test articles and other NASA instrumentation/equipment as necessary, including during pre-test build up and at other times when not actively undergoing testing (to include while awaiting their shipment offsite post-test).
- Provide transportation of NASA ESS pack test articles from the offerors test lab storage facility to test site and back
- Provide mounting interface for mounting ESS module test articles to drop trolley
- Instrument test articles and drop mechanism prior to test in accordance with a mutually agreed upon instrumentation plan
- Provide test instrumentation for any of the following as needed: accelerometers, strain gauges, thermocouples, photogrammetric speckle coat patterns, load cells, or other TBD instrumentation to supplement any instrumentation supplied by NASA.
- Provide recording and storage capabilities at least 20 kHz for the above listed instrumentation for the duration of the drop test.

- Provide high speed video (min 1000 fps), from at least 4 sides, infrared video and real time video for each drop test.
- Provide low speed (100 Hz or less) monitoring capabilities of instrumentation and infrared video for each ESS module test article for up to one hour post-test.
- Provide all safety items associated with the testing. At minimum, safety items include, but are not limited to:
 - A 50-gallon minimum freshwater dunk tank, sufficient submerging the test articles
 - A secure outside storage facility such as a CONEX container capable of storing and monitoring the test articles and instrumentation post-test until their shipment offsite post-test.
 - A fire suppression system capable of extinguishing the test articles on the test stand should a fire initiate
 - Methods to safely and securely transfer test articles between the drop facility/trolley and dunk tank and storage facility
 - Onsite fire mitigation support through local firefighting personnel

Engineering Support:

- Work with NASA to develop final test plan which includes the identification of instrumentation, test matrix, test collection methods, hazard analysis and other test specific items
- Work with NASA to help identify test specific items such as test article implementation into the test apparatus, instrumentation layout, test site planning, or other technological or logistical issues that arise during test preparations

Removal:

- Package and ship tested ESS test articles in accordance with all relevant Department of Transportation (DOT) regulations for Li-Ion batteries and/or Hazardous Materials to a location of NASA's choosing for posttest inspections. Ship all other instrumentation back to NASA Langley Research Center.

3. Deliverables:

- Provide to NASA raw data files of all instrumentation data collected for all tests, in the form of American Standard Code for Information Interchange (ASCII) or another mutually agreed upon format
- Provide to NASA raw video files collected from all tests, in a mutually agreed upon format
- Provide test execution checklists for all tests

4. Period of Performance:

This contract will expire June 30, 2026